

OpenPrescribing in R

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Overview

This tutorial walks you through how to work with the [OpenPrescribing.net](#) BigQuery data in R. It will cover:

1. **Connect to the BigQuery database from R** using [DBI](#)¹ and [bigrquery](#)²
2. **Description of the BigQuery schema** using [dm](#)³
3. **Reusable functions to work with the data** using [dbplyr](#)⁴

1 Connect to BigQuery

1.1 Authorise BigQuery

To get access to the OpenPrescribing BigQuery database write a message in the #openprescribing Slack channel and tag slack-admin. One of the admins will send you a JSON file with your BigQuery credentials in a private message. The file should look something like this:

```
{
  "type": "ACCOUNT_TYPE",
  "project_id": "PROJECT_ID",
  "private_key_id": "PRIVATE_KEY_ID",
  "private_key": "PRIVATE_KEY",
  "client_email": "CLIENT_EMAIL",
  "client_id": "CLIENT_ID",
  "auth_uri": "AUTH_URI",
  "token_uri": "TOKEN_URI",
  "auth_provider_x509_cert_url": "AUTH_CERTIFICATE",
  "client_x509_cert_url": "CLIENT_CERTIFICATE",
  "universe_domain": "DOMAIN"
}
```

IMPORTANT: Never commit this file to GitHub or share it with anyone.

1.1.1 Where to store your credentials

You can store this file anywhere on your computer, but a common location is in a hidden folder in your home directory, for example:

On macOS/Linux:

```
~/.config/openprescribing/bq-credentials.json
```

On Windows:

```
C:/Users/YOUR_USERNAME/.config/openprescribing/bq-credentials.json
```

1.1.2 Set up the environment variable

Rather than hardcoding the path in your R scripts you can set an environment variable named `OP_CREDENTIALS` or similar. To help you edit your `.Renviron` file and set the variable for all R sessions you can use the `usethis` package:

```
# Open your .Renviron file
usethis::edit_r_environ()
```

This will open your `.Renviron` file in your editor where you can add the location where you stored your credentials file, for example:

```
# macOS/Linux
OP_CREDENTIALS=~/.config/openprescribing/bq-credentials.json

# Windows
OP_CREDENTIALS=C:/Users/YOUR_USERNAME/.config/openprescribing/bq-credentials.json
```

Note: You will probably need to restart your R session for R to know about your new environment variable.

1.1.3 Connecting to BigQuery

The following code connects to the `hscic` dataset in the `ebmdatalab` project. You will need to authenticate using the path to your credentials file stored in the `OP_CREDENTIALS` environment variable and might be asked to login with your Google account the first time you run this.

```
library(DBI)
library(bigrquery)

bq_auth(path = Sys.getenv("OP_CREDENTIALS"))

con <- dbConnect(
  bigrquery::bigrquery(),
  project = "ebmdatalab",
  dataset = "hscic",
  billing = "ebmdatalab"
)
```

2 Database schema

2.1 Main hscic tables

This has multiple tables, we're now looking at the main ones we use for research projects using the OpenPrescribing data. To see all the tables available in the `hscic` dataset, you can use:

```
dbListTables(con)
```

2.1.1 normalised_prescribing

Add description here.

2.1.2 regional_teams

Add description here.

2.1.3 practices

Add description here.

2.1.4 practice_statistics

Add description here.

2.2 Relationship diagram?

Add description here.

Columns		
sha	<i>character</i>	Strategic Health Authority code
regional_team	<i>character</i>	Regional team identifier
stp	<i>character</i>	Sustainability and Transformation Partnership code
pct	<i>character</i>	Primary Care Trust code
practice	<i>character</i>	Practice code
bnf_code	<i>character</i>	British National Formulary (BNF) code
bnf_name	<i>character</i>	British National Formulary (BNF) drug name
items	<i>integer</i>	Count of prescription items
net_cost	<i>numeric</i>	Net ingredient cost
actual_cost	<i>numeric</i>	Actual cost?
quantity	<i>numeric</i>	Total quantity prescribed?
month	<i>POSIXct</i>	Prescribing month

Rows 10 Columns 12

Columns		
code	<i>character</i>	Regional team code
name	<i>character</i>	Regional team name
open_date	<i>Date</i>	Open date
close_date	<i>Date</i>	Closing date, NA if still open

Rows 10 Columns 4

Columns		
code	<i>character</i>	Practice code
name	<i>character</i>	Practice name
address1	<i>character</i>	Address line 1
address2	<i>character</i>	Address line 2
address3	<i>character</i>	Address line 3
address4	<i>character</i>	Address line 4 (town/city)
address5	<i>character</i>	Address line 5
postcode	<i>character</i>	Practice postcode
location	<i>character</i>	Geographic location?
cgc_id	<i>character</i>	Clinical Commissioning Group identifier
pcn_id	<i>character</i>	Primary Care Network identifier
setting	<i>integer</i>	?
close_date	<i>character</i>	Date practice closed
join_provider_date	<i>character</i>	Date practice joined WHAT?
leave_provider_date	<i>character</i>	Date practice left WHAT?
open_date	<i>character</i>	Date practice opened
status_code	<i>character</i>	Practice status code

Rows 10 Columns 17

Columns		
month	<i>POSIXct</i>	The month for which practice statistics are recorded
male_0_4	<i>integer</i>	Count of male patients aged 0-4 years
female_0_4	<i>integer</i>	Count of female patients aged 0-4 years
male_5_14	<i>integer</i>	Count of male patients aged 5-14 years
male_15_24	<i>integer</i>	Count of male patients aged 15-24 years
male_25_34	<i>integer</i>	Count of male patients aged 25-34 years
male_35_44	<i>integer</i>	Count of male patients aged 35-44 years
male_45_54	<i>integer</i>	Count of male patients aged 45-54 years
male_55_64	<i>integer</i>	Count of male patients aged 55-64 years
male_65_74	<i>integer</i>	Count of male patients aged 65-74 years
male_75_plus	<i>integer</i>	Count of male patients aged 75 years and over
female_5_14	<i>integer</i>	Count of female patients aged 5-14 years
female_15_24	<i>integer</i>	Count of female patients aged 15-24 years
female_25_34	<i>integer</i>	Count of female patients aged 25-34 years
female_35_44	<i>integer</i>	Count of female patients aged 35-44 years
female_45_54	<i>integer</i>	Count of female patients aged 45-54 years
female_55_64	<i>integer</i>	Count of female patients aged 55-64 years
female_65_74	<i>integer</i>	Count of female patients aged 65-74 years
female_75_plus	<i>integer</i>	Count of female patients aged 75 years and over
total_list_size	<i>integer</i>	Total Count of patients registered at the practice
astro_pu_cost	<i>numeric</i>	ASTRO PU? cost
astro_pu_items	<i>numeric</i>	ASTRO PU? items
star_pu	<i>character</i>	STAR PU?
pct_id	<i>character</i>	PCT? identifier
practice	<i>character</i>	Practice code
Rows 10 Columns 25		

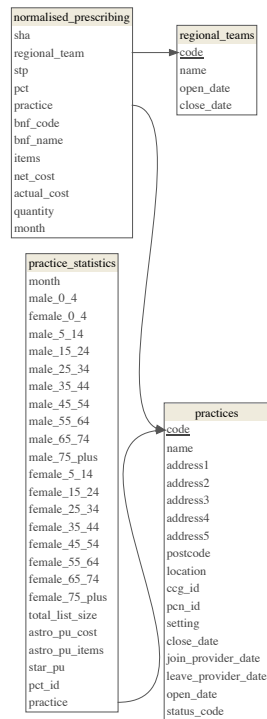


Figure 2.1: Relationship diagram for hscic tables

3 Reusable functions

Write some functions here using `dbplyr`⁴.

4 Applied example

Replicate some work here that illustrates how nice it is to do this in R.

References

1. R Special Interest Group on Databases (R-SIG-DB), Wickham, H. & Müller, K. *DBI: R Database Interface*. (2024). doi:[10.32614/CRAN.package.DBI](https://doi.org/10.32614/CRAN.package.DBI).
2. Wickham, H. & Bryan, J. *bigrquery: An Interface to Google's 'BigQuery' 'API'*. (2025). doi:[10.32614/CRAN.package.bigrquery](https://doi.org/10.32614/CRAN.package.bigrquery).
3. Schieferdecker, T., Müller, K. & Bergant, D. *dm: Relational Data Models*. (2025). doi:[10.32614/CRAN.package.dm](https://doi.org/10.32614/CRAN.package.dm).
4. Wickham, H., Girlich, M. & Ruiz, E. *dbplyr: A 'Dplyr' Back End for Databases*. (2024). doi:[10.32614/CRAN.package.dbplyr](https://doi.org/10.32614/CRAN.package.dbplyr).